

Requirements Document

Project Title: KU-Based Transfer Credit Matchmaking System

Team: Data Science Team

Stakeholders: Faculty evaluators, academic advisors, transfer students, CAE institutions

1. Introduction

1.1 Purpose

The purpose of this system is to automate and standardize the evaluation of transfer credits between CAE (Centers of Academic Excellence) institutions using **Knowledge Unit (KU) mapping** rather than relying solely on course titles or one-to-one equivalencies.

The system will assist academic evaluators by recommending course equivalencies based on KU coverage while providing transparent evidence and flags for human review.

1.2 Problem Statement

Manual transfer credit evaluation is:

- Time-consuming
- Inconsistent across reviewers
- Difficult when courses differ in structure or titles
- Hard to scale as transfer volume increases

Since courses across institutions often cover the same **KUs** but in different combinations, a KU-based matchmaking system can provide a more flexible and accurate method of determining transfer eligibility.

1.3 Project Goals

The system should:

1. Map transfer courses to target program requirements based on KU overlap
 2. Recommend equivalencies using an explainable scoring method
 3. Identify partial matches and gaps
 4. Respect institutional policies (credit hours, level, etc.)
 5. Provide evidence to support decisions
 6. Support human review and overrides
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2. Scope

2.1 In Scope

The system will:

- Accept a student's transfer course list
- Identify source institution and target institution/program
- Retrieve KU coverage for courses
- Compare source KU coverage to target requirements
- Use an algorithmic approach to recommend course equivalencies
- Output:
 - Accepted transfer matches
 - Partial matches
 - Missing requirements
 - Explanation of KU coverage
- Flag cases needing human review

2.2 Out of Scope (for this phase)

- Full automation without human review
- Direct transcript parsing from PDFs (assume structured input)
- Legal/financial transfer policy processing
- Real-time integration with university SIS systems
- Automatic KU extraction from raw syllabi (future work)

3. Stakeholders

Stakeholder	Role
Academic Advisors	Use system recommendations to assist students
Faculty Evaluators	Review/approve transfer equivalencies
Transfer Students	Benefit from faster decisions
CAE Institutions	Provide KU mapping data
Development Team	Builds and maintains the system

4. System Overview

The system functions as a **decision-support tool**, not a final authority.

High-level workflow:

1. Student transfer data is input
 2. System retrieves KU mappings for source courses
 3. System retrieves KU requirements for target program
 4. Matching algorithm evaluates coverage
 5. System proposes equivalencies and gaps
 6. Human reviewer confirms or overrides
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5. Functional Requirements

FR1 – Input Transfer Course Data

The system shall accept a structured list of transfer courses including:

- Course title
 - Institution
 - Credits
 - Level (if available)
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FR2 – KU Retrieval

The system shall retrieve the set of KUs associated with each course from a KU database.

FR3 – Target Requirement Retrieval

The system shall retrieve KU requirements for:

- Target courses
 - Target program requirement groups
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FR4 – KU Coverage Scoring

The system shall compute a KU coverage score between source courses and target requirements using:

$$Coverage = \frac{|KU_{source} \cap KU_{target}|}{|KU_{target}|}$$

The system may weight KUs based on importance (core vs elective).

FR5 – Matching Algorithm

The system shall determine the best mapping between source courses and target requirements using a constrained optimization or greedy set cover approach.

FR6 – Policy Constraints

The system shall enforce institutional policies including:

- Minimum credit hours
 - Course level compatibility
 - Grade thresholds (if provided)
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FR7 – Output Recommendations

The system shall output:

- **Accepted equivalencies**
 - **Partial matches**
 - **Unmet requirements**
 - **Confidence / review flags**
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FR8 – Explanation Generation

For each recommendation, the system shall display:

- Matched KUs
 - Missing KUs (if partial)
 - Policy checks
 - Evidence references (course descriptions, KU notes)
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FR9 – Human Review Support

The system shall allow reviewers to:

- Approve recommendation
 - Override recommendation
 - Add notes for future reference
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6. Non-Functional Requirements

Category	Requirement
Explainability	All decisions must include KU-based reasoning
Accuracy	System should prioritize minimizing false positive acceptances
Usability	Output must be understandable to non-technical faculty
Scalability	Should support multiple institutions and programs
Extensibility	Must allow future addition of ML components
Transparency	Must clearly flag uncertain cases

7. Data Requirements

The system requires:

- Institution database
 - Program and course catalogs
 - Course $\rightarrow$ KU mappings
 - Program requirement $\rightarrow$ KU mappings
 - Policy rules per institution
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8. Assumptions

- KU mappings exist or can be curated
 - Transfer input is structured
 - Faculty will remain part of decision process
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9. Future Enhancements (Not in MVP)

- ML-based KU prediction from syllabus text
- NLP matching for course title similarity
- Confidence scoring models
- Direct integration with institutional systems

10. Success Criteria

The project will be considered successful if:

- The system produces reasonable equivalency recommendations
- Faculty reviewers find explanations useful
- The system reduces manual evaluation time
- The system correctly identifies most KU overlaps